

Name: Niel Jhon E. Celocia**Date: Jul 28, 2026****Course & Year: BSIT-4****Remarks: _____****ACTIVITY:****1. What is the Angular Ecosystem, and what are its main components?**

The Angular Ecosystem is a complete collection of tools, libraries, and frameworks designed for building scalable web applications. Its main components include the **Angular Framework** (core architecture like components, services, and routing), **Angular CLI** (command-line tool), **RxJS** (reactive programming library), **TypeScript** (the primary language), and **Angular Material** (UI component library).

2. What is Angular CLI, and why is it important in Angular development?

Angular CLI (Command Line Interface) is an official command-line tool used to initialize, develop, scaffold, test, and maintain Angular applications. It is important because it automates setup, enforces standard project structure, streamlines code generation, and simplifies building and bundling code.

3. What is Node.js, and why is it required for Angular?

Node.js is an open-source, cross-platform JavaScript runtime environment that allows JavaScript to execute outside a web browser. It is required for Angular because it powers the package manager (**npm**) used to install Angular dependencies and runs the development server and build tools locally.

4. How do Angular CLI and Node.js work together when developing Angular applications?

Node.js provides the runtime environment and package management (npm) that hosts Angular CLI. When you run an Angular CLI command (like **ng serve** or **ng build**), Node.js executes the underlying JavaScript tasks to process, bundle, and serve your application.

5. Why is it important for Angular developers to understand these three topics before building applications?

Understanding these three core areas gives developers a clear picture of how project setup, dependency management, build processes, and runtime tooling interact. This foundation makes project configuration smoother, simplifies debugging environment issues, and leads to cleaner, well-structured application development.